

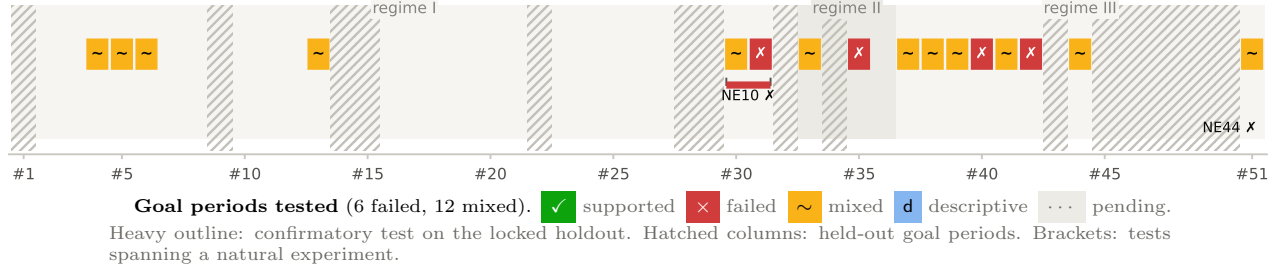
H30: An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Operator messages have a measurable period-level susceptibility, about one extra active minute per nudge for its target (twice round 1's value once dropped events are restored) and a small content pull for human messages, but day-to-day steerability is not a trackable state.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

How much does one operator message move the agents who see it, and can that number be tracked day by day? 16 non-holdout goal periods, after a synthetic validation that fixed three estimator biases. **Round 1b** re-ran everything on the corrected activity table (round 1 lost half of all events), with the nudge's leading @ as its only target, kicks timed at the call that reads them, and past-only controls with a day fixed effect. *Activity*: a nudge adds about one active minute for its target in both large periods (G51 0.98 [0.63, 1.37], G38 0.95 [0.38, 1.54]; round 1: 0.59 and -0.08). First nudges 1.21, repeats 0.44; room-mates ≈ 0 . Round 1's "don't nudge into a lull" reverses (lull 1.74 vs 0.72). *Content*: human messages turn recipients' next statements toward their new content, +0.024 (room-mates) and +0.041 (named), in both embedding models. *Daily gauge*: some day-to-day variation in G51 ($p = 0.026$, $R_1 = 0.48$) but no persistence. Natives: nudges buy more under the 5-min pause default than the 12-h one (NE44, reversed). Holdout not run.

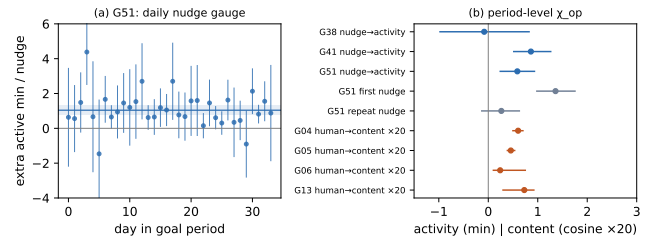
PHYSICAL MODEL

Models 01 and 11: linear response to impulsive fields. Each operator message k is a field pulse on its recipients i . The activity spin is $n_i(t) \in \{0, 1\}$. Content state: a whitened statement embedding s_i . Message direction u_k , projected off the span of the recipient's last 8 statements ($u_k^\perp$). χ_{act} : the 30-min integrated activity response, from a local projection with matched pre-history strata α_s , day effects δ_d and kick counts K . χ_{con} : post-message alignment relative to same-kind placebo messages p . $\chi_{\text{op}}(d)$: the day mean of per-kick responses r_k .

$$Y_i(m) \equiv \sum_{\tau=1}^{30} n_i(m + \tau) = \alpha_{s(i,m)} + \delta_d + \sum_c \chi_{\text{act}}^c K_i^c(m) + \beta \cdot K_i^{\text{adj}}(m) + \varepsilon$$

$$\chi_{\text{con},k} = \bar{s}_i^{\text{post}} \cdot u_k^\perp - \langle \bar{s}_i^{\text{post}} \cdot u_p^\perp \rangle_p, \quad u^\perp = \frac{(1 - P_{\text{pre}})u}{\|(1 - P_{\text{pre}})u\|}$$

$$\chi_{\text{op}}(d) = \frac{\sum_{k \in d} w_k r_k}{\sum_{k \in d} w_k}, \quad R_1 = 1 - \frac{\text{Var}_{\text{perm}}[\chi_{\text{op}}(d)]}{\text{Var}[\chi_{\text{op}}(d)]}$$



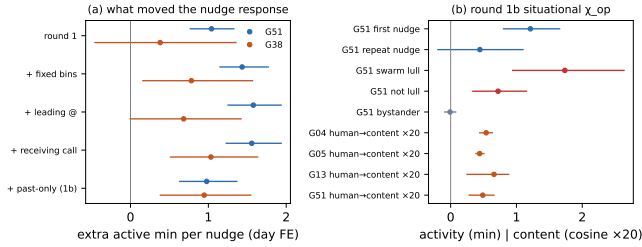
Observables. (a) The G51 daily nudge gauge: extra active minutes per nudge for the target, with day fixed effect, 95% CIs, and the period mean and band. Day-to-day scatter is about the size of the daily CIs: no trend, no persistence. (b) Period-level χ_{op} : nudge→activity in the three largest nudge periods (pre-registered model); first vs repeat nudges in G51 (day FE); human→content in the human-dense periods (cosine ×20).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful as a set of corrections rather than as a gauge. Someone steering an agent swarm should not watch a daily "steerability" number: at village message rates it is a period constant plus noise, and it does not age. What does carry over: one nudge to an idle agent buys about a minute of extra work, and only the first one does. Repeats and nudges sent into a swarm-wide lull buy almost nothing, and room-mates ignore other agents' nudges. To change *what* agents write, say something new and name the agent; that moves content, not activity. The methods lesson may matter more: conditioning on whether the nudger fires again silently biased H04's estimate. The effects are small, single-embedding, and unconfirmed on the holdout.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict
P1 nudge→target	G51 0.59 → 0.98 [0.63, 1.37]; G38 -0.08 → 0.95 [0.38, 1.54]; small periods 5/10 > 0	✓ (count rule ×)
P2 bystanders ≈ 0	G51 0.13 → -0.01; G38 -0.54 → -0.09	✓
P3 human→activity	≤ 0 / n.s. in regime I (unchanged)	×
P4 human→content	0.026 → 0.024 (gte 0.024); named 0.046 → 0.041	✓
P5 nudge→content	G51 0.003 → 0.002 [-0.001, 0.005]	✓
P6 no daily state	G51 p 0.08 → 0.026, R_1 0.28 → 0.48, lag-1 0.08	× as written
P7 no aging	G51 +0.02 [-0.03, 0.07]/day	✓
P8 context sawtooth	early 0.66 vs late 1.49	×
P10 family	no Holm-significant lab	✓
P11 nulls	G51 and G38 now pass	✓
Native #5 dose	pull falls 0.025→0.013 over 1–7 msgsg; 8+ noisy	× (rule)
Native NE44	12-h 0.57 vs 5-min 0.98 (Δ -0.41)	× reversed
Native NE10	first nudges ever -0.42 (22 kicks)	× low power



Second figure. Round 1b. (a) What moved the nudge response in G51 (blue) and G38 (orange): restoring the dropped events carries most of the change; past-only controls bring the level back to ≈ 1 min. (b) Situational χ_{op} in G51 (first vs repeat nudges, swarm lull vs not, bystanders) and human content pull ($\times 20$).

GOAL PERIODS IN DETAIL

G04 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G05 $\sim$ 1b: mixed (r1 mixed) · native dilution: failed
 G06 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G13 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G30 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G31 $\times$ 1b: failed (r1 mixed; fixed bins, receiving call)
 G33 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G35 $\times$ 1b: failed (r1 failed; fixed bins, receiving call)
 G37 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G38 $\sim$ 1b: mixed (r1 mixed; fixed bins, receiving call)
 +8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only G51 and G38 resolve the nudge response; 3–5-day periods give unstable CIs. The level depends on the future-kick adjustment (0.98 past-only vs 1.56 with round 1's A1 regressors, day FE). SE-based window reliabilities exceed 0.5 in G51 but also in G38, where the calibrated permutation finds no day signal, so they are not counted. Human activity in dense regime-I chat is not identified. Content measures message-specific steering only; human replies may share topics. Round 1b's design (receiving-call timing, past-only, day FE primary) follows the re-evaluation brief, not the pre-registration.

ROUND 2

- **Essence.** a trustworthy answer to "will my next message change what this swarm does?". Round 1 says it is per situation, not a daily state.
- **H30-R1** Situational χ instead of daily χ .
- **H30-R2** The 2026-08-20 switch-off as a natural experiment.
- **H30-R3** Embedding swap and reply control for χ_{con} .
- **H30-R4** Per-message content strength.
- **H30-R5** Run confirm.py after the card is committed.

H30-operator-susceptibility · generated 2026-10-04 09:20 by infra/summaries/build_summaries.py · card hypotheses/H30-operator-susceptibility/README.md · ratings: H30 round-1b agent (suggested) (page 2)